Base station physics: summary

Params: base/spec/params.py v0.1-stub (still defaults for LAUNCH, AIM, REEL, LAYOUT)

Spring (each, A228, SF 1.2 on Shigley static limits)

Rate used / by formula	0.125 / 0.131 N/mm
Initial tension used	7.0 N
Body coils, mass	96, 54 g
Safe force (governed by)	43.6 N (hook torsion)
Safe extension (wire) / spec cap	293 / 250 mm
Surge (fixed-fixed)	25 Hz

Launch (chain spring model)

Stroke, ratio, pitch	245 mm, 1.0:1, 35 deg
Spring extension full / rest	280 / 35 mm
Stored energy (2 springs)	13.1 J
Carriage force at full draw	84 N
Muzzle speed	7.61 m/s
Peak acceleration, stroke time	21 g, 55 ms
Efficiency (dart KE / stored)	62 %
Carriage KE at stop, pad peak	2.32 J, 434 N (15 mm pad)
Spring KE at stop, coil clash	0.94 J, 8.3 m/s

Flight (bead-chain tether + reel model, level ground)

Range at 0.5 / 0.75 / 1.0 N brake	4.93 / 4.46 / 4.04 m (pitch 36)
Same shot, no tether	5.93 m
Flight time, apex (0.75 N)	0.89 s, 1.24 m
Impact speed, angle (0.75 N)	6.2 m/s, 54 deg down
Ballistics table	6 to 47 deg, best 36.0 deg: 4.45 m

Storage reel payout (0.75 N at full reel)

Reel empty mass, inertia full	35 g, 19.4 kg mm2
Line radius full / core	27.4 / 20.0 mm (brake +37 % at core)
Line pull in the stroke (spin-up)	4.37 N
Reel rpm max	2652 rpm
Overrun after landing	0.25 m in 0.13 s

Capstan and tow

Wraps for tow limit (mu 0.15)	3.3 raw, 5 design
Take-up needed (design wraps)	0.45 N
Tow at limit: torque, speed, current	2.02 N m, 0.119 m/s, 3.72 A
Line pull at limiter (5 N m) (12.6 V)	127 N (break 667 N)
Line pull, motor stall (no limiter, 12.6 V)	358 N

Drivetrain (wiper at 11.1 V with sag)

aim: peak torque, current, time	0.33 N m, 1.9 A, 7 s
cock: peak torque, current, time	3.07 N m, 4.9 A, 3 s
reel: peak torque, current, time	0.17 N m, 1.7 A, 142 s
tow: peak torque, current, time	2.02 N m, 3.7 A, 42 s
Wiper at limiter (5 N m), 12.6 V (with sag)	5.0 N m, 7.0 A
Wiper motor stall (no limiter), 12.6 V (with sag)	14.1 N m, 17.0 A

Harry must measure (then edit the parameter block and rerun: python summary.py)

- Spring hang test: hang 1, 2, 3 kg from one spring, measure the length each time. Slope = rate, intercept = initial tension. Every range number scales with the stored energy.
- Wiper: free speed (turns in 30 s at 12.0 V), free current, stall torque (arm + luggage scale, 1 s) and stall current. Edit common.Wiper or spec WIPER.
- Dart mass on a kitchen scale with ballast and tether knot (spec 0.279 kg).
- Slip clutch: pull the line off the full reel with a luggage scale; set 0.4 to 0.5 N (FINDINGS F5).
- Rail friction: tilt the rail with the dart on it until it slides (mu = tan angle).